01:XXX:XXX - Homework n

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1. Question 1.

   Solution: This is my solution

   Testing

2. Question 2.

   Solution:

3. Question 3.

   Solution:

4. Question 4. Prove that in an arbitry topological space, the union of two closed sets is closed, using the limit point definition of closed sets.

   Solution: Suppose A and B are closed sets in a topological space (X, τ). Let $C = A \cup B$. We want to show that C is closed, which means we need to show that C contains all its limit points.

   Suppose for contradiction that there exists a limit point x of C such that $x \notin C$.

   Since x is a limit point of C, every open neighborhood of x must intersect C in some point other than x.

   Since $C = A \cup B$, this means that every open neighborhood of x must intersect either A or B (or both) in some point other than x. But since A and B are closed, they contain all their limit points. This is a contradiction.

5. Question 5.

   Solution: